Introduction to Software Engineering Project– HW 1

Gameplay Example:

The inputs I entered in order:

1 -> 2 -> 3 -> 4 -> 4 -> 3

The outputs and the game situation in each player decision:

```
choose an action:
 1 - attack the enemy
 2 - defend
 3 - special
 4 - recover 1
1
Player HP: 90 Enemy HP: 20 Player Resource: 5
choose an action:
 1 - attack the enemy
 2 - defend
 3 - special
 4 - recover 2
2
Player HP: 85 Enemy HP: 20 Player Resource: 5
choose an action:
 1 - attack the enemy
 2 - defend
 3 - special
 4 - recover 3
3
Player HP: 75 Enemy HP: 5 Player Resource: 2
choose an action:
 1 - attack the enemy
 2 - defend
 3 - special
 4 - recover 4
4
Player HP: 65 Enemy HP: 5 Player Resource: 5
choose an action:
 1 - attack the enemy
 2 - defend
 3 - special
 4 - recover 4
4
Player HP: 55 Enemy HP: 5 Player Resource: 8
choose an action:
 1 - attack the enemy
 2 - defend
 3 - special
 4 - recover 3
3
Player HP: 55 Enemy HP: 0 Player Resource: 5
You win
```

Explaining the outputs:

Each loop I printed:

1. The options the player can play
2. His decision
3. The situation after the player and the enemy played

After the enemy/player died , I stopped the loop and printed if the player won or not

Test and results:

Before every test the starting point is:

Player hp = 100

Enemy hp = 30

Player resources = 5 (not includes W01-RES-01)

test	input	expected	Actual
W01-ATK-01	1	Enemy hp = 20	Enemy hp = 20
W01-DEF-01	2	Player hp = 95	Player hp = 95
W01-SPC-01	3	Player resources = 2 Enemy hp = 15	Player resources = 2 Enemy hp = 15

W01-RES-01	I changed a bit the code that the player will start with 9 resources by:
	this.resource = 9;
	// Only for W01-RES-01 test
	Then entered:4
		Player resources = 10	Player resources = 10

W01-INV-01	hi	Player hp = 100 Enemy hp = 30 Player resources = 5	Player hp = 100 Enemy hp = 30 Player resources = 5

Python vs Java Comparison

python	java
You don't have to declare the type a variable will be or what type a function will return	You must declare the type a variable will be and what type a method will return
To enter a condition / loop / function/ … you put the code that you want to be there line under and using indentation.	To enter a condition / loop / function/ … you put the code that you want to be there inside {}.
You can terminate a statement without putting anything at the end of the line.	Most of the statements must end with ";"
If there isn't a class and a main function, the starting point will be in the first line.	The starting point must be in a function public static void main(String [] args) inside of a class. If there isn't one you won't be able to run the code.
The code runs through an interpreter which may compile it to bytecode internally.	The code compiles to bytecode and then runs through JVM.

Design Note

The main design decision I made was to divide my game into three separate classes: Main, player_class, and enemy_class.

Firstly, I decided to separate the player and enemy from Main because I thought it would put too much responsibility on Main. I also wanted to declare variables as private so I could keep things like HP and resources private. Therefore, I made the first split, separating Main from the characters.

Then I saw in the guide that the player would have a few different elements than the enemy, and I understood that it would be better to put the player and the enemy into two separate classes because of their different methods and the fact that eventually we would like the enemy to move/act differently from the player. For these reasons, I decided to split the player and the enemy into two classes.

This decision made it much easier for me to find errors, connect the characters, and use methods. One downside I can think of is that I have to create a method like getHP() so Main can access the player's HP because it's private.

Engineering quality and AI task

Prompt summary - I asked the AI to review my Java game code and identify possible logical errors or edge cases based on the assignment instructions. I asked it to provide at least two testable suggestions, explain the problems, and give input sequences to test them. I also asked it not to change my code or suggest features outside the assignment requirements.

Suggestion 1:

Potential problem: A player might attempt a Special attack without enough resources, but the game could incorrectly allow an enemy counterattack.

Test:

I entered those inputs in this order: 3 -> 3 -> 1

The values in the beginning

Player hp = 100

Enemy hp = 30

Resources = 5

Input	What should happen	What happened
3	Player hp = 90 Enemy hp = 15 Resources = 2	Player hp = 90 Enemy hp = 15 Resources = 2
3	Player hp = 90 Enemy hp = 15 Resources =2	Player hp = 90 Enemy hp = 15 Resources =2
1	Player hp = 80 Enemy hp = 5 Resources =2	Player hp = 80 Enemy hp = 5 Resources =2

Suggestion 2:

Potential problem: Recover might allow resources to exceed 10 or might incorrectly cause a counterattack when the player is already at maximum resource.

Test:

I entered those inputs in this order: 4 -> 4 -> 4

The values in the beginning

Player hp = 100

Enemy hp = 30

Resources = 5

Input	What should happen	What happened
4	Player hp = 90 Enemy hp = 30 Resources = 8	Player hp = 90 Enemy hp = 30 Resources = 8
4	Player hp = 80 Enemy hp = 30 Resources =10	Player hp = 80 Enemy hp = 30 Resources =10
4	Player hp = 80 Enemy hp = 30 Resources =10	Player hp = 80 Enemy hp = 30 Resources =10

Conclusion:

I tested both AI suggestions using the input sequences above that the AI gave me. In both cases, the game worked as expected, so I decided not to change my code. The suggestions were still useful because they helped me verify that my game follows the assignment rules.

Known Limitations: No known limitations